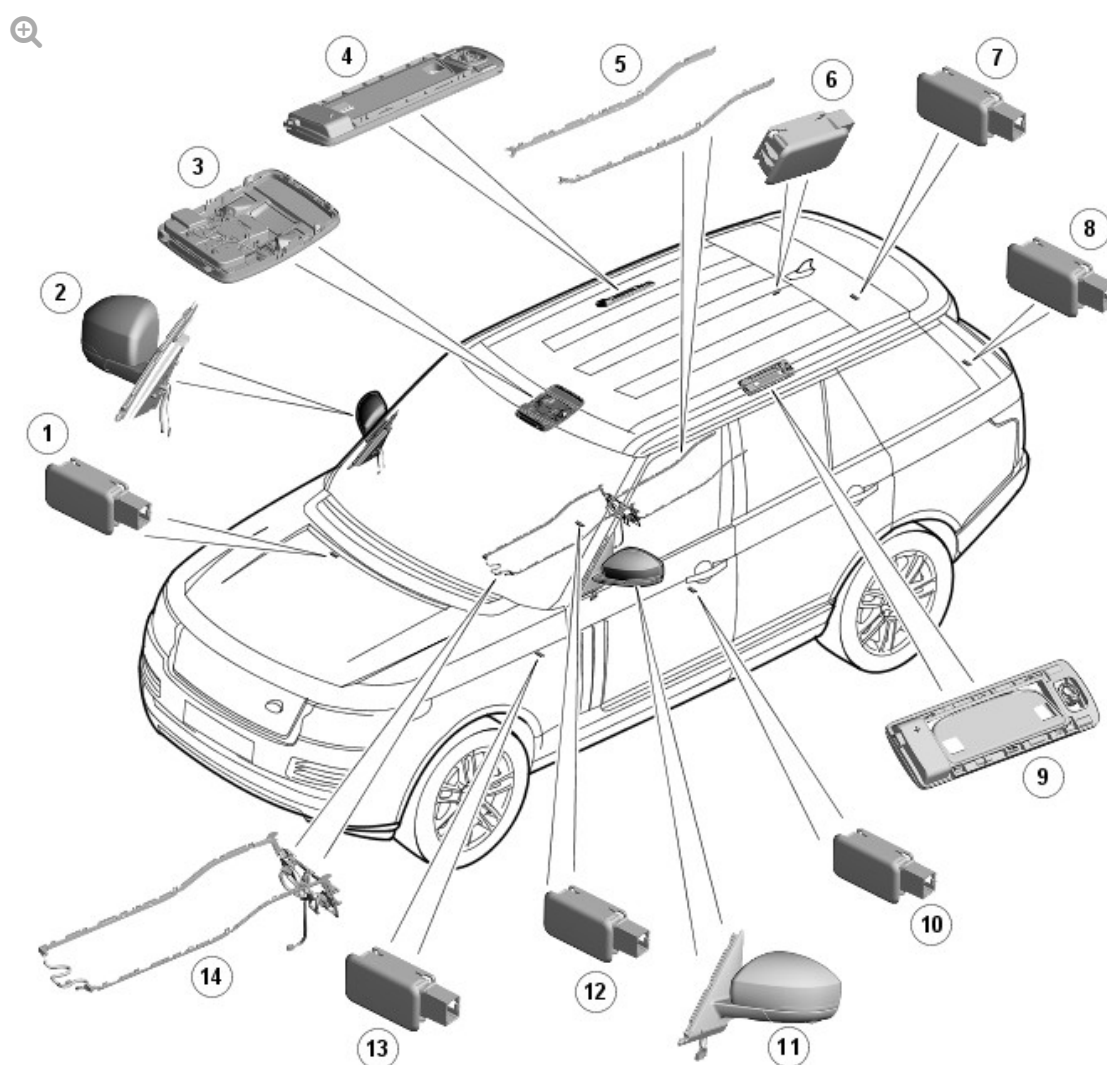


2016.0 RANGE ROVER (LG), 417-02

INTERIOR LIGHTING

DESCRIPTION AND OPERATION

COMPONENT LOCATION

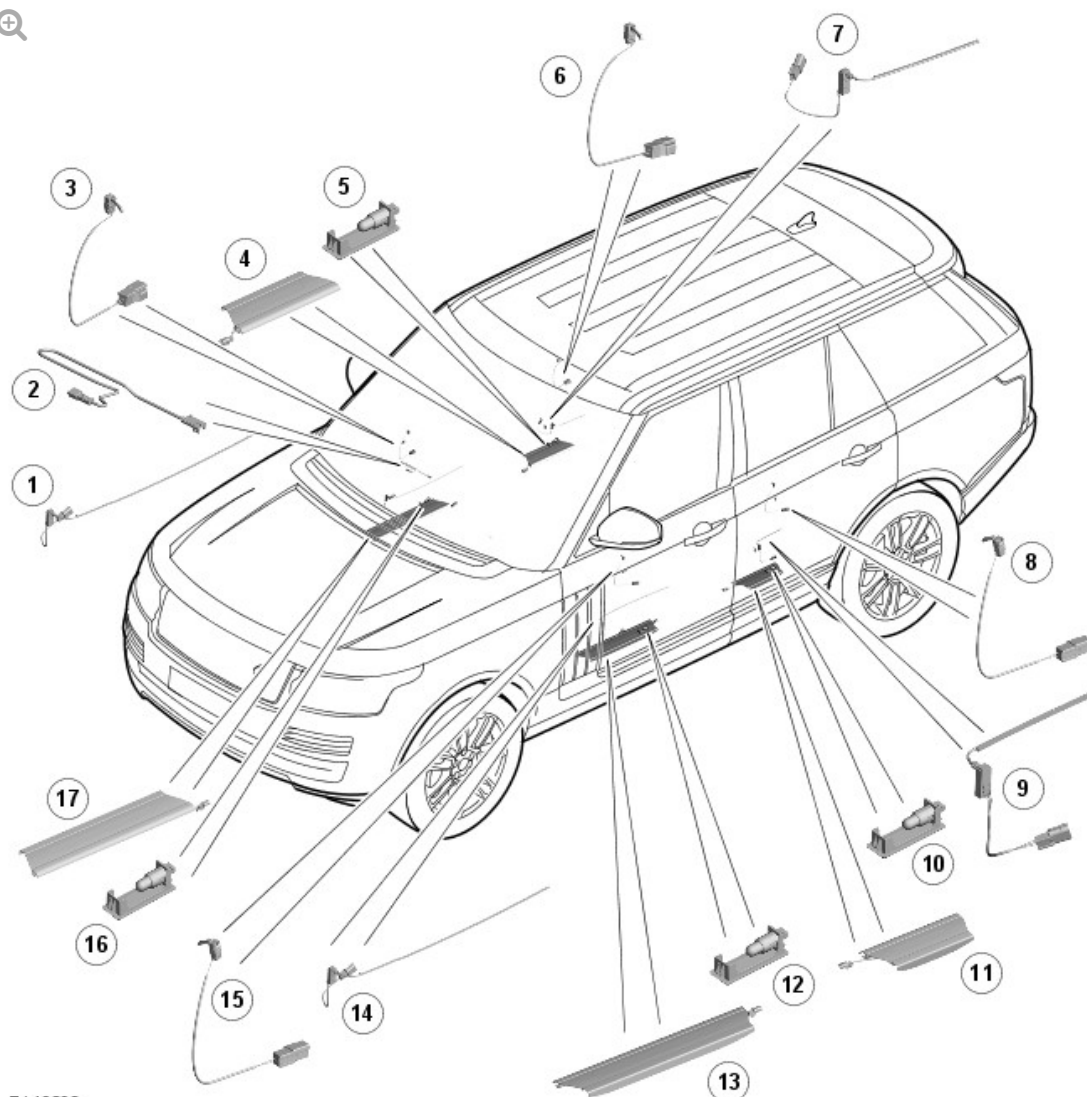


E149491

ITEM

DESCRIPTION

1	Front right footwell lamp
2	Right side approach lamp
3	Front interior lamp
4	Rear right interior lamp
5	Rear ambience lighting (Business class seats only)
6	Load space lamp
7	Tailgate load space lamp
8	Tailgate load space lamp
9	Rear left interior lamp
10	Rear left footwell lamp
11	Left side approach lamp
12	Rear right footwell lamp
13	Front left footwell lamp
14	Front ambience lighting



E149293

ITEM	DESCRIPTION
1	Front right door pocket illumination
2	Glove compartment switch/lamp
3	Front right door handle illumination
4	Rear right tread plate illumination
5	Rear right door puddle lamp
6	Rear right door handle illumination
7	Rear right door pocket illumination
8	Rear left door handle illumination
9	Rear left door pocket illumination
10	Rear left door puddle lamp

11	Rear left tread plate illumination
12	Front right door puddle lamp
13	Front left tread plate illumination
14	Front left door pocket illumination
15	Front left door handle illumination
16	Front right door puddle lamp
17	Front right door tread plate illumination

OVERVIEW

NOTES:

- Unless otherwise stated the interior lighting system uses LED (light emitting diode)s instead of bulbs.
- The LED illumination of the front door pockets and the front and rear interior door handles is a function of the CJB (central junction box) and only operates when the light control switch is in the side lamp or headlamp positions.

The interior lighting is controlled by the CJB which is located behind the glove compartment. The interior lighting comprises the following lamps:

- Front interior lamp unit
- Two rear interior lamp units
- Upper tail door load space lamps (2 off)
- Luggage compartment load space lamp
- Treadplate lamps (4 off)
- Under fascia footwell lamps (2 off)
- Rear passenger footwell lamps (2 off) located in the seat base of the front

seats

- Door mirror approach lamps (2 off)
- Puddle lamps (4 off).

The interior lamps have two modes of operation; automatic or manual. In the automatic mode the interior lamp functionality is controlled by the CJB on receipt of various input signals. In manual mode, the lamps can be switched on and off using the momentary button adjacent to the front interior lamp or can be disabled completely using the same switch.

The following table shows the bulbs used for the interior lighting and their type and rating:

BULB	TYPE	RATING
Door puddle lamps	Capless	5W

DESCRIPTION

FRONT AND REAR INTERIOR LAMPS AND MAP READING LAMPS

The front and rear interior lamp assemblies are similar in design. The front interior lamp switch is connected directly to the CJB and controls the manual operation of the interior lamp functionality. The rear interior lamp switch is used only to switch the lamp on and off.

The map reading lights have a non-adjustable beam and are controlled via separate switches.

LUGGAGE COMPARTMENT LAMPS

The luggage compartment lamps comprise one lamp unit located on the RH (right-hand) side trim panel, adjacent to the auxiliary power socket, and two lamps located in the interior trim panel of the upper tail door. All of these lamps are of the same design. The lamps are controlled by the tail door latch assembly and the CJB and operate on the automatic interior lamp functionality.

GLOVE COMPARTMENT LAMP

The glove compartment lamp is located inside the glove compartment, at the top, adjacent to the latch. The lamp housing also incorporates a plunger type switch which is activated to switch the lamp on when the glove compartment lid is opened.

VANITY MIRROR ILLUMINATION LAMPS

The vanity mirror lamps are located in the front sun visors. The lamps are located behind a hinged cover, either side of the mirror glass. The lamps are activated by a simple switch which operates when the mirror cover is lifted.

FOOTWELL LAMPS

Two footwell lamps are located in the lower fascia closing panels and illuminate the area below the fascia. The lamps are controlled by the CJB and operate on the automatic interior lamp functionality.

DOOR PUDDLE LAMPS

Each of the doors is fitted with a puddle lamp to illuminate the ground below the door when the door is open. The front door puddle lamps are located in the lower face of the door, facing the ground. The rear door puddle lamps are located in the bottom of the door trim panel.

The puddle lamps are controlled by the CJB and operate on the automatic interior lamp functionality.

DOOR MIRROR APPROACH LAMPS

The exterior mirror housings are fitted with a lamp which illuminates the area below the front door. The housing contains a plastic lens which directs a beam of light in the required area.

The lamps are controlled by the CJB, via the front driver and passenger door modules. The CJB sends a CAN (controller area network) message to the door module to illuminate the approach lamps when an unlock signal, a front door open signal or an ignition off mode is detected.

DIAGNOSTICS

Diagnostics for the interior lamps is limited to a check of the function of the front interior lamp switch. The CJB can be interrogated using a Land Rover approved diagnostic system to establish the correct operation of the front interior lamp switch, the door switches and the tail door switch.

OPERATION

INTERIOR LIGHTING

The interior lamp operation is activated by any of the following input signals:

- Door switch contacts (front and rear)
- Tail door latch assembly
- Power Mode 4
- Central locking, lock request
- Central locking remote handset operation
- Crash sensor
- Front interior lamp switch.

The load space lamp and the upper tail door lamps are not switched on with the interior lamps. These lamps are only activated when the upper tail door is opened. When the upper tail door is opened, the interior lamps, in addition to the load space lamps, are switched on and are subject to the CJB timers. The tail door open signal to the CJB is provided by a microswitch in the tail door latch assembly.

AUTOMATIC OPERATION

The automatic control of the interior lamps has several switch on and off conditions as follows:

- The interior lamps are switched on when one or more doors are opened. When all the doors are closed, the interior lamps are switched off after a delay of 20 seconds. If the vehicle is locked after closing the doors, within

the 20 second period, the lamps go off immediately.

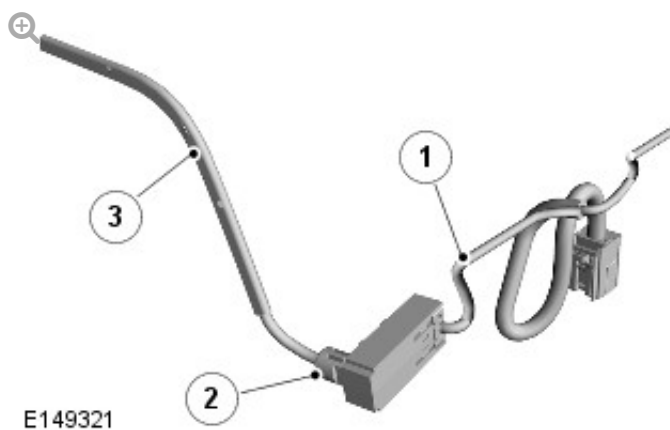
- When the interior lamps are on and the ignition is in power mode 0 or 6, the lamps go off immediately after all doors have been closed.
- In power mode 0, the interior lamps are switched on for 20 seconds. This timer is overridden if a door is subsequently opened.
- When the vehicle is unlocked and the doors are closed, the interior lamps can be activated for a further 20 second period by pressing the 'unlock' button on the remote handset.
- When the vehicle is unlocked from a locked condition, the interior lamps are activated for a 20 second period. This timer is overridden if a door is subsequently opened.
- If a door is left open, the interior lamps will remain on for 16 minutes. After this period, the CJB timer switches the interior lamps off to prevent excessive battery drain.

The CJB uses a PWM (pulse width modulation) operation of the interior lamps for the on/off conditions. The lamps are switched on with a 'soft ramp' of 1.3 seconds. The lamps are switched off with a 'soft ramp' of 2.6 seconds. The soft on/off operation is controlled by a semiconductor within the CJB

LED Ambience Lighting Component

NOTE:

Example shown, units vary with application.



ITEM	DESCRIPTION
1	Electrical supply
2	LED light source
3	LED Light Guide

The ambience lighting systems are controlled by the CJB. Customer ambient lighting selections in the TS (touch screen) are processed and sent to the CJB via the MS (medium speed) CAN comfort bus. The CJB output voltage to the LED's is varied via PWM. This function influences the lighting intensity providing the combined lighting effect.

MANUAL OPERATION

The interior lamps can be switched on or off using the manual switch for the front interior lamp. The front interior lamp switch will switch all interior lamps, front and rear, on or off. If the interior lamps are on when the switch is pressed, all interior lamps will go off. The on and off automatic switching conditions remain effective even when the manual switch is used.

In power mode 4 or 6 and the front interior lamp switch is used to switch the interior lamps on, the lamps will remain on continuously. If the ignition is put into power mode 0, the interior lamps will be subject to the 16 minute CJB timer. The interior lamp is immediately switched off when the vehicle is locked. After a locking procedure, the interior lamp automatic control is reinstated.

The rear interior lamp switch will switch the rear interior lamp only on or off.

Continuous Off

To prevent drain on the battery if the doors are to remain open for some time, the interior lamps can be turned off permanently. This is stored in the CJB EEPROM and is also activated following a system power down or a reset of the CJB

To enter the continuous off condition, the front interior lamp switch must be pressed and held for more than 3 seconds. The continuous off function can only be disabled by pressing the front interior lamp switch, after which, full automatic control is resumed. When continuous off is active, the rear interior lamp switch can still be used to switch the rear interior lamp only on or off.

CRASH SIGNAL OPERATION

The CJB receives a crash signal on a hardwired connection from the RCM (restraints control module) in the event of a crash of a severity to activate the airbags. On receipt of this signal, the CJB activates the interior lamps which are on permanently and not subject to the CJB timer.

The crash operation of the interior lamps can only be cancelled by operating the manual switch on the front interior lamp or by locking and unlocking the vehicle.

AMBIENCE LIGHTING

The ambience lighting system has variable colour options which can be adjusted by the customer using a menu option in the Touch-screen.

The ambience lighting system uses phosphor blue and red LED's which are combined together to make a range of five different colours from pure blue, through purple, to pure red. In addition to the colour of the ambient lighting, customers can also vary the brightness.

Automatic colour changes are also activated when the driver selects dynamic mode; the ambience lighting automatically changes from the default blue colour to red (this colour change is also repeated in the

instrument cluster graphics).

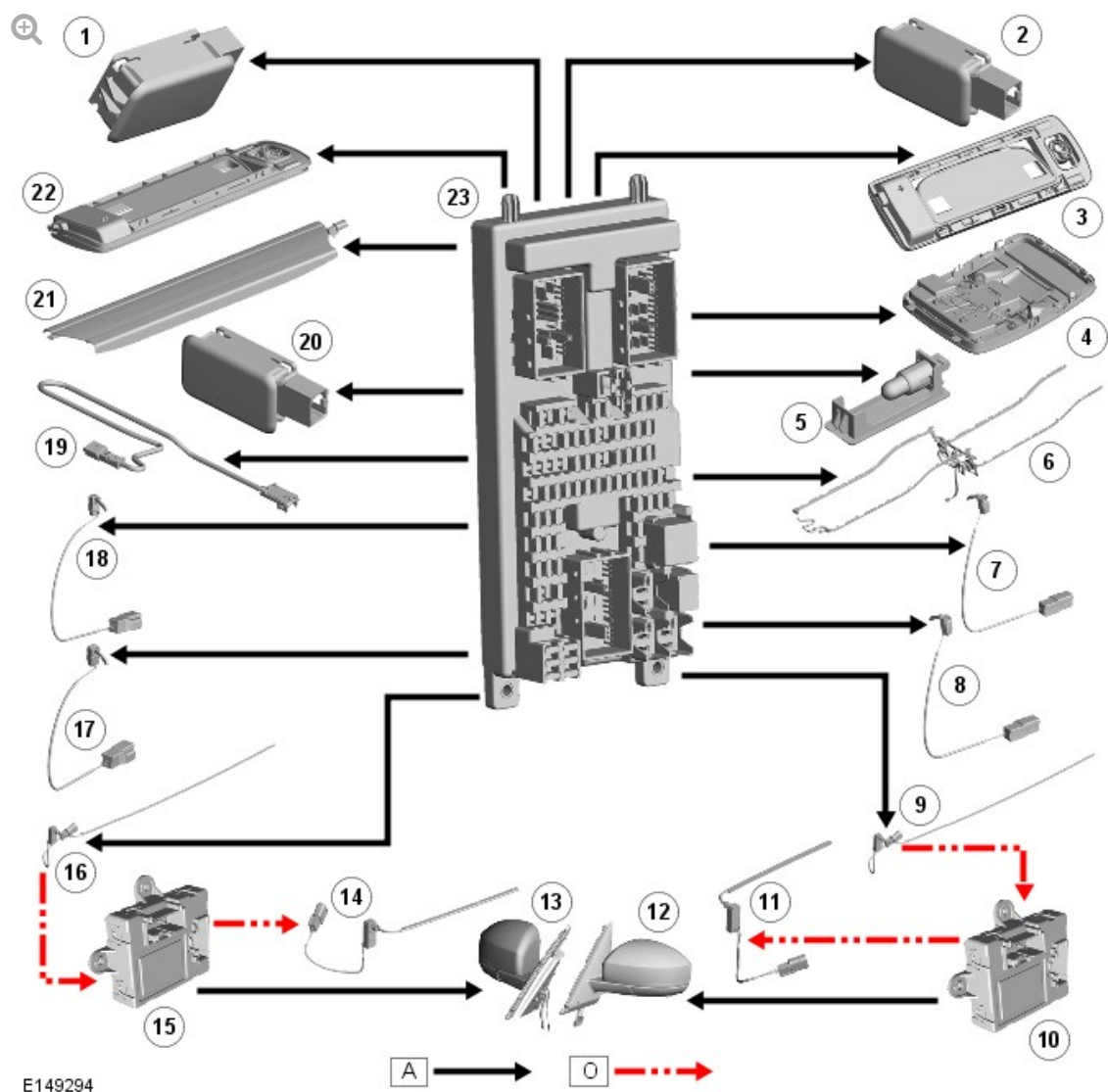
The LED ambience lighting is located:

- In the door grab handles
- In the door pockets
- Along the centre console edges (including the rear business class centre console)

Ambience lighting is provided by LED units around the vehicle. Each component is made up of three parts:

- LED light source
- A light guide
- A light guide carrier

CONTROL DIAGRAM



A = HARDWIRED; O = LIN (LOCAL INTERCONNECT NETWORK) BUS.

ITEM	DESCRIPTION
1	Load space lamp
2	Tailgate load space lamp x2
3	Rear right interior lamp
4	Front interior lamp
5	Door puddle lamps x4
6	Centre console ambience lighting
7	Rear right door handle illumination
8	Rear right door handle illumination

9	Front right door pocket illumination
10	Driver Door Module (DDM)
11	Rear right door pocket illumination
12	Right approach lamp
13	Left approach lamp
14	Rear left door pocket illumination
15	Passenger Door Module (PDM)
16	Front left door pocket illumination
17	Rear left door handle illumination
18	Front left door handle illumination
19	Glove compartment switch/lamp
20	Footwell lamps (4 Off)
21	Tread plate illumination x4
22	Rear right interior lamp
23	Central Junction Box (CJB)